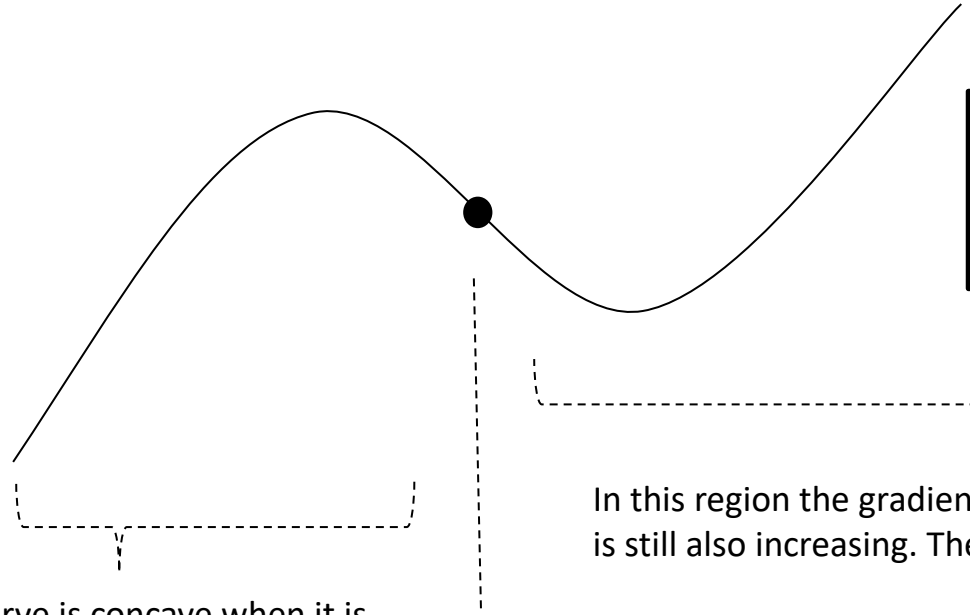
P2 Chapter 9: Differentiation

Using The Second Derivative

Using the second derivative

Year 1 slides you may remember that a point of inflection was where the concavity of a curve changes, i.e. concave to convex or vice versa, or informally, 'swerving one way to swerving the other'.



Memory tip:

Concave = hill
Convex = valley

In this region the gradient, although initially negative, is still also increasing. The curve is **swerving left**.

We say the curve is concave when it is **swerving right**: we can see the gradient is always decreasing, i.e. the change in the gradient is negative.

At point of inflection, the gradient is not changing, i.e. the gradient of the gradient is 0. Note that points of inflection (as with this example) are not necessarily stationary points.

✎ $f(x)$ convex when $f''(x) \geq 0$

✎ $f(x)$ concave when $f''(x) \leq 0$

✎ Point of inflection when $f''(x) = 0$

Examples

[Textbook] Find the interval on which the function $f(x) = x^3 + 4x + 3$ is concave.

?

[Textbook] Show that $f(x) = e^{2x} + x^2$ is convex for all real values of x.

?

[Textbook] The curve C has equation $y = x^3 - 2x^2 - 4x + 5$
Find the coordinates of the point of inflection.

?

Examples

[Textbook] Find the interval on which the function $f(x) = x^3 + 4x + 3$ is concave.

$$f'(x) = 3x^2 + 4$$
$$f''(x) = 6x$$
Concave $\therefore 6x \leq 0$
$$x \leq 0$$

[Textbook] Show that $f(x) = e^{2x} + x^2$ is convex for all real values of x.

$$f'(x) = 2e^{2x} + 2x$$
$$f''(x) = 4e^{2x} + 2$$
$e^{2x} > 0$ for all $x \in \mathbb{R}$
$\therefore f''(x) > 0$ for all $x \in \mathbb{R}$ therefore convex for all $x \in \mathbb{R}$

[Textbook] The curve C has equation $y = x^3 - 2x^2 - 4x + 5$
Find the coordinates of the point of inflection.

$$\frac{dy}{dx} = 3x^2 - 4x - 4 \qquad \frac{d^2y}{dx^2} = 6x - 4$$
Point of inflection $\therefore 6x - 4 = 0$
$$x = \frac{2}{3}$$
$$\therefore y = \left(\frac{2}{3}\right)^3 - 2\left(\frac{2}{3}\right)^2 - 4\left(\frac{2}{3}\right) + 5 = \frac{47}{27}$$
$$\left(\frac{2}{3}, \frac{47}{27}\right)$$

Exercise 9.9

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Homework Exercise

1 For each of the following functions, find the interval on which the function is:

i convex **ii** concave

a $f(x) = x^3 - 3x^2 + x - 2$ **b** $f(x) = x^4 - 3x^3 + 2x - 1$ **c** $f(x) = \sin x,\ 0 < x < 2\pi$

d $f(x) = -x^2 + 3x - 7$ **e** $f(x) = e^x - x^2$ **f** $f(x) = \ln x,\ x > 0$

2 $f(x) = \arcsin x,\ -1 < x < 1$

a Show that $f'(x) = \dfrac{1}{\sqrt{1 - x^2}}$

b Hence show that $f(x)$ is concave on the interval $(-1, 0)$.

c Show that $f(x)$ is convex on the interval $(0, 1)$.

d Hence deduce the point of inflection of f.

3 Find any point(s) of inflection of the following functions.

a $f(x) = \cos^2 x - 2\sin x,\ 0 < x < 2\pi$ **b** $f(x) = -\dfrac{x^3 - 2x^2 + x - 1}{x - 2},\ x \neq 2$

c $f(x) = \dfrac{x^3}{x^2 - 4},\ x \neq \pm 2$

4 $f(x) = 2x^2 \ln x,\ x > 0$

Show that f has exactly one point of inflection and determine the value of x at this point.

5 The curve C has equation $y = e^x(x^2 - 2x + 2)$.

a Find the exact coordinates of the stationary point on C and determine its nature.

b Find the coordinates of any non-stationary points of inflection on C.

Homework Exercise

6 The curve C has equation $y = xe^x$.

a Find the exact coordinates of the stationary point on C and determine its nature.

b Find the coordinates of any non-stationary points of inflection on C.

c Hence sketch the graph of $y = xe^x$.

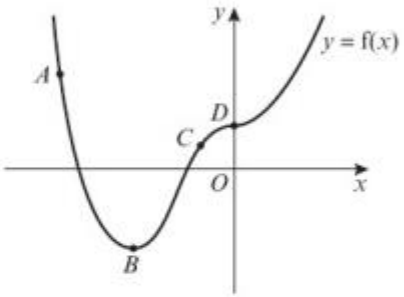
Problem-solving

Consider how C behaves for very large positive and negative values of x.

7 For each point on the graph, state whether:

i $f'(x)$ is positive, negative or zero

ii $f''(x)$ is positive, negative or zero



8 $f(x) = \tan x, -\frac{\pi}{2} < x < \frac{\pi}{2}$

Prove that $f(x)$ has exactly one point of inflection, at the origin.

9 Given that $y = x(3x - 1)^5$,

a find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ **(4 marks)**

b find the points of inflection of y. **(4 marks)**

Homework Exercise

10 A student is attempting to find the points of inflection on the curve C with equation

$$y = (x - 5)^4$$

The attempt is shown below:

$$\frac{dy}{dx} = 4(x - 5)^3$$

$$\frac{d^2y}{dx^2} = 12(x - 5)^2$$

When $\frac{d^2y}{dx^2} = 0$,

$$12(x - 5)^2 = 0$$

$$(x - 5)^2 = 0$$

$$x - 5 = 0$$

$$x = 5$$

Therefore, the curve C has a point of inflection at $x = 5$.

a Identify the mistake made by the student. **(2 marks)**

b Write down the coordinates of the stationary point on C and determine its nature. **(2 marks)**

11 A curve C has equation

$$y = \frac{1}{3}x^2 \ln x - 2x + 5, \; x > 0$$

Show that the curve C is convex for all $x \geqslant e^{-\frac{3}{2}}$. **(5 marks)**

Challenge

1 Prove that every cubic curve has exactly one point of inflection.

2 The curve C has equation $y = ax^4 + bx^3 + cx^2 + dx + e, \; a \neq 0$

a Show that C has at most two points of inflection.

b Prove that if $3b^2 < 8ac$, then C has no points of inflection.

Homework Answers

1 **a** **i** $[1, \infty)$ **ii** $(-\infty, 1]$

b **i** $(-\infty, 0] \cup \left[\frac{3}{2}\ \infty\right)$ **ii** $\left[0, \frac{3}{2}\right]$

c **i** $[\pi, 2\pi)$ **ii** $(0, \pi]$

d **i** nowhere **ii** $(-\infty, \infty)$

e **i** $[\ln 2, \infty)$ **ii** $(-\infty, \ln 2]$

f **i** nowhere **ii** $(0, \infty)$

2 **a** Let $y = f(x)$. Then $x = \sin y$.

$$\frac{dx}{dy} = \cos y \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - \sin^2 y}}$$

so $f'(x) = \dfrac{1}{\sqrt{1 - x^2}}$

b $f'(x) = \dfrac{1}{\sqrt{1 - x^2}}$, $f''(x) = \dfrac{x}{(1 - x^2)^{\frac{3}{2}}}$

$f''(x) \leqslant 0 \Rightarrow x \leqslant 0$, so $f(x)$ concave for $x \in (-1, 0)$

c $f''(x) \geqslant 0 \Rightarrow x \geqslant 0$, so $f(x)$ convex for $x \in (0, 1)$

d $(0, 0)$

3 **a** $\left(\frac{\pi}{6}, -\frac{1}{4}\right), \left(\frac{5\pi}{6}, -\frac{1}{4}\right)$

b $(1, -1)$ **c** $(0, 0)$

4 $f'(x) = 2x + 4x \ln x = 2x(1 + 2 \ln x)$, $f''(x) = 6 + 4 \ln x$

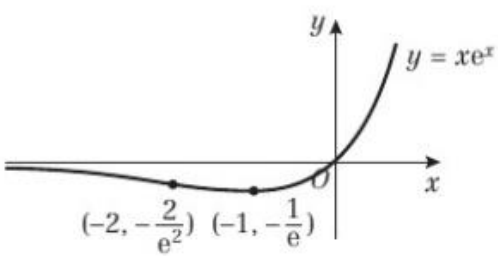
$f''(x) = 0 \Rightarrow 4 \ln x = -6 \Rightarrow \ln x = -\frac{3}{2} \Rightarrow x = e^{-\frac{3}{2}}$

There is one point of inflection where $x = e^{-\frac{3}{2}}$

5 **a** $(0, 2)$, point of inflection **b** $\left(-2, \dfrac{10}{e^2}\right)$

6 **a** $\left(-1, -\frac{1}{e}\right)$, minimum **b** $\left(-2, -\frac{2}{e^2}\right)$

c



8 $f'(x) = \sec^2 x$, $f''(x) = 2 \sin x \sec^3 x$

$f''(x) = 0 \Leftrightarrow \sin x = 0$ (as $\sec x \neq 0$) $\Leftrightarrow x = 0$

So there is one point of inflection at $(0, \tan 0) = (0, 0)$.

9 **a** $\dfrac{dy}{dx} = 15x(3x - 1)^4 + (3x - 1)^5$

$\dfrac{d^2y}{dx^2} = 30(3x - 1)^4 + 180x(3x - 1)^3$

b $\left(\frac{1}{9}, -\frac{32}{2187}\right), \left(\frac{1}{3}, 0\right)$

10 **a** Although $\dfrac{d^2y}{dx^2} = 0$, the sign does not change, so there is not a point of inflection when $x = 5$.

b $(5, 0)$; minimum

11 $\dfrac{dy}{dx} = \dfrac{2}{3}x \ln x + \dfrac{1}{3}x - 2$, $\dfrac{d^2y}{dx^2} = \dfrac{2}{3} \ln x + 1$

$\dfrac{d^2y}{dx^2} \geqslant 0 \Leftrightarrow \dfrac{2}{3} \ln x \geqslant -1 \Leftrightarrow x \geqslant e^{-\frac{3}{2}}$

Challenge

1 A general cubic can be written as $f(x) = ax^3 + bx^2 + cx + d$.

$f''(x) = 6ax + 2b$. $f''(x) = 0 \Leftrightarrow x = -\dfrac{b}{3a}$

Let $\varepsilon > \mathbb{R}$, $\varepsilon > 0$:

$f''\left(-\dfrac{b}{3a} + \varepsilon\right) = 6a\varepsilon > 0$, $f''\left(-\dfrac{b}{3a} - \varepsilon\right) = -6a\varepsilon < 0$

So the sign of $f''(x)$ changes either side of $x = -\dfrac{b}{3a}$, and this is a point of inflection.

2 **a** $f''(x) = 12ax^2 + 6bx + 2c$ is quadratic, so there are at most two values of x at which $f''(x) = 0$.

b $\dfrac{d^2y}{dx^2} = 12ax^2 + 6bx + 2c$

Discriminant $= 36b^2 - 96ac < 0 \Leftrightarrow 3b^2 < 8ac$

So when $3b^2 < 8ac$, $\dfrac{d^2y}{dx^2} = 0$ has no solutions.

Therefore C has no points of inflection.